Proving Ground

Make anything.
Watch a real robot deal with it.



Aholo Lux3D turns a photo or a sentence into a physical object. Unitree's published G1 walking controller has to get past it, live in the browser.

proving-ground-snowy.vercel.app

Manycore Aholo Global 3D AI Hackathon · Rohan Borade (solo) · November 2026

Robots fail on the things nobody modelled

Walking and grasping controllers are tested on a handful of hand-made shapes. Real floors have shoes, toys, cables, chairs and bags.

Lux3D builds almost any object at its real size from a sentence or a photo: about 90 seconds of generation, and about 2 minutes until it stands in front of the robot.

So: a robot test range where every obstacle is made by Lux3D, and the robot is a real controller, not an animation.

Robotics teams

Varied, realistic test objects with real sizes and collision parts in a few minutes, downloadable as MuJoCo-ready packs.

Anyone curious about robots

Make something from your own life and watch a real controller deal with it. Share the course that beat it.

Lux3D as an asset factory

Shows Lux3D output going straight into a physics simulator: size, parts, people and motion, no hand modelling.

MEASURED, NOT ESTIMATED

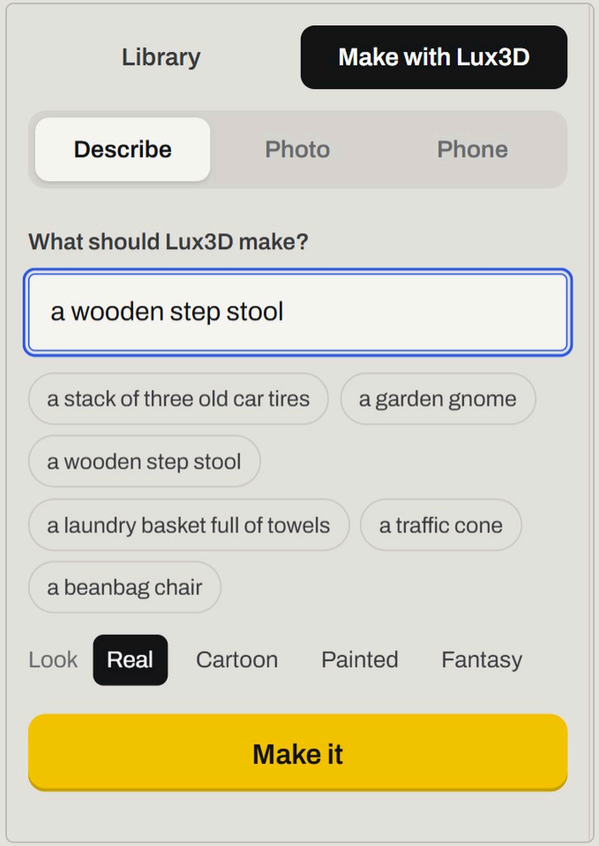
79–147 s
Lux3D generation per object
Median 89 s over 66 library generations.

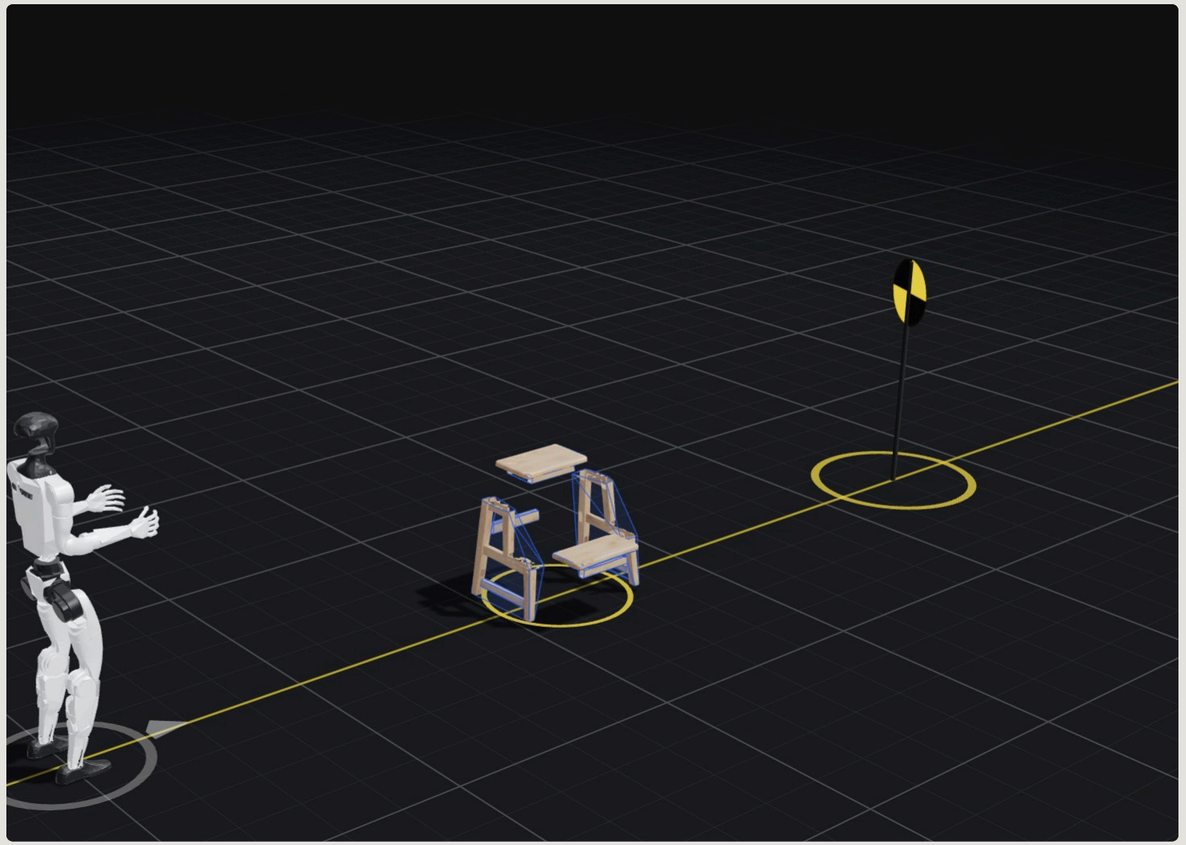
~2 · ~4 min
A finished object, end to end
Single part, and with part-split (a toolbox took 114 s, a folding chair 208 s).

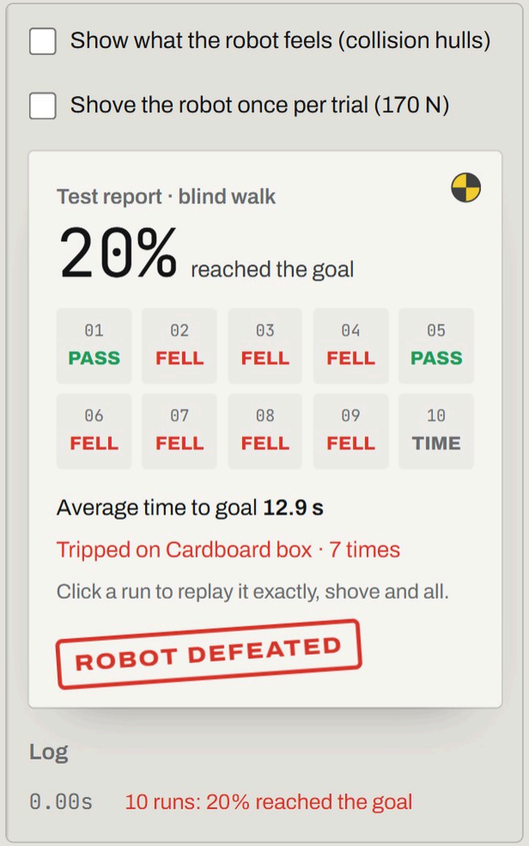
2.5 µm
Gap to Unitree's own Python run
Pelvis position after 2 s of walking.

12×
Real time in the browser
One chair; 3–4× with twenty objects in the scene.

From a sentence to a test result in three steps







1 Make it

Type a sentence, upload a photo, or scan the QR code and snap with your phone. Lux3D builds it in about 90 seconds.

2 Drop it in

It materializes at its real size in front of the robot: shape, texture, parts, then the hulls the physics feels. Drag it, turn it, change its density.

3 Test it

Walk once, steer, or run ten seeded trials for a pass rate, the falls, and what it tripped on. Save the course and share the link.

Screenshots from the arena. The test report shown is a blind walk on the starter course (box, chair, a Lux3D-made jogger crossing): 2 of 10 runs reached the flag.

Every object passes through Lux3D

Step	Aholo call	Credits	Time
Upload a photo or GLB	Asset upload (OUS)	0	1–3 s
Photo to 3D	img-to-3d · G1-Turbo	1	~90 s
Sentence to 3D	text-to-3d · G1-Turbo	13	~90 s
Split into parts	part-split	30	~95 s
Rig a person	animations/rig	15	~63 s
Make them walk	animations/retarget · Walk_Loop	6	~120 s
Hinges for doors and gates	articulation · URDF	28	~3 min
STL / OBJ / FBX download	multi-format-export	0	~30 s
Price every job first	pricing/openapi-quotes	0	<1 s

Library snapshot, Nov 9, 2026, 07:13 IST, all made by Lux3D

72 objects and people

231 collision parts

30 split by part-split

6 rigged walkers

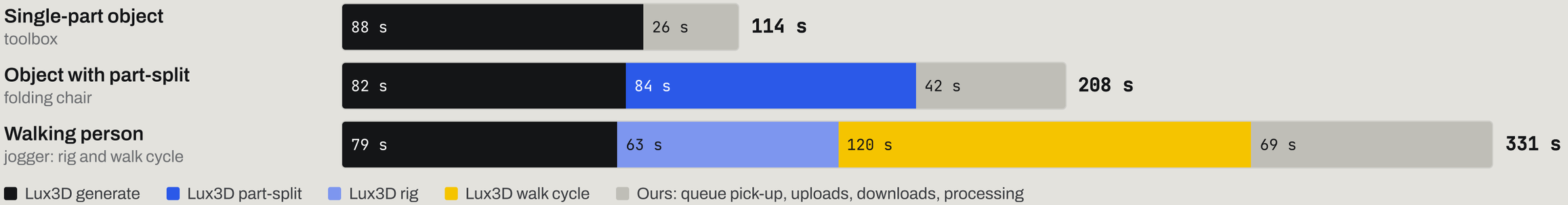
Inputs must go through Aholo's own upload: third-party image URLs fail silently, so every photo and every intermediate GLB is re-uploaded.

Lux3D runs one task per account, so a single queue serves everyone; the library keeps the site playable while it is busy.

Lux3D predicts real size for most objects: a sneaker comes back 28 cm long and a chair 85 cm tall with no one typing a dimension. When a size is more than 35% off the typical one for that kind of thing (a street trash can came back 38 cm tall), the pipeline rescales it.

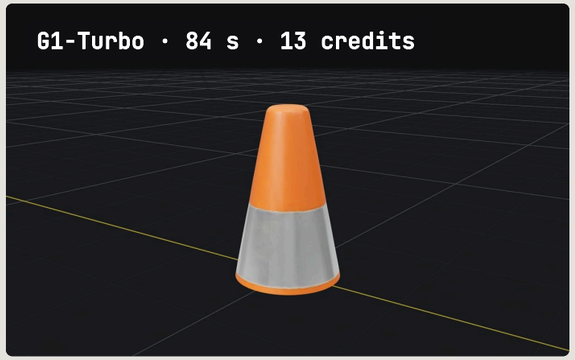
Where the time goes

seconds from the stored records of three finished jobs



Iteration 1: Lux3D Turbo against G1

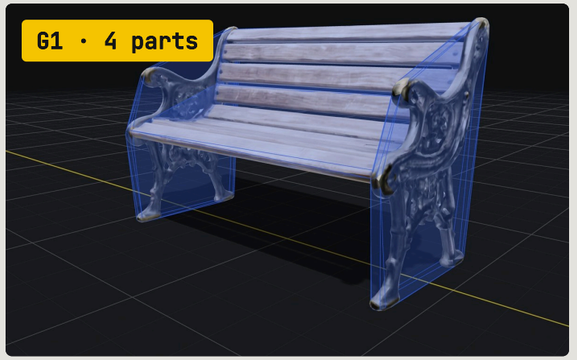
Traffic cone · “an orange traffic cone with a white reflective band”





Park bench · “a wooden park bench with cast iron legs” · blue = part hulls





	Cone · Turbo	Cone · G1	Bench · Turbo	Bench · G1
Generation	84 s	285 s	85 s	295 s
Total, end to end	88 s	288 s	179 s	382 s
Credits	13	20	43 (13 + 30)	50 (20 + 30)
Triangles from Lux3D	18,210	7,586	19,294	21,210
GLB from Lux3D	7.9 MB	4.0 MB	10.6 MB	8.2 MB
File the browser loads	313 KB	150 KB	685 KB	455 KB
Parts	1	1	5	4

Same prompts, same sizes (0.7 m cone, 1.8 m bench). G1 took 3.4× as long and 1.5× the credits for the cone, and it draws the white band and orange base like a real cone where Turbo's band comes back grey. A 90 s wait is what lets someone make an object and test it in one sitting, so Turbo stays the default and G1 is one setting (LUX_VERSION=G1).

Iteration 2: one hull, or one per part



The chair is a solid, slanted block: the gaps between the legs do not exist.

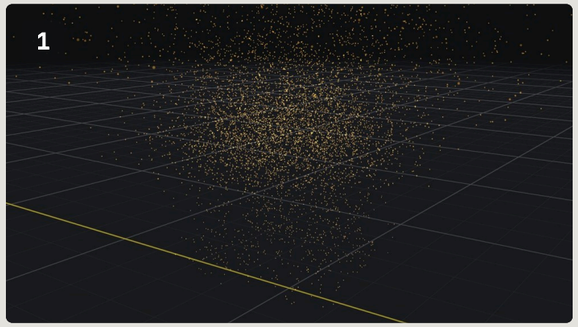


Lux3D part-split: legs, rungs, seat and back are separate solids with their own mass.

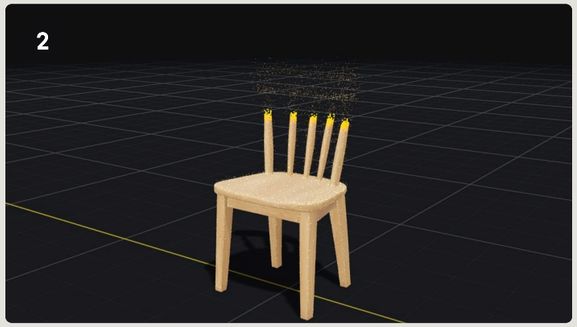
	First try	Now
Collision	1 hull for the chair	1 hull per part (8)
Chair mass	18.8 kg (hull volume)	about 5 kg (mesh volume)
Reached the flag, 20 walks	14	18
Mean time to the flag	16.7 s	10.9 s

20 blind walks per version (two seeds of ten), chair square-on 1.5 m ahead. Most of the gap is mass: the single hull at the corrected mass gets 9 of 10 in 11.4 s. The parts add shape: the eight part hulls together fill 49.8 L for 12.9 L of chair, so legs, seat and back each act as their own solid.

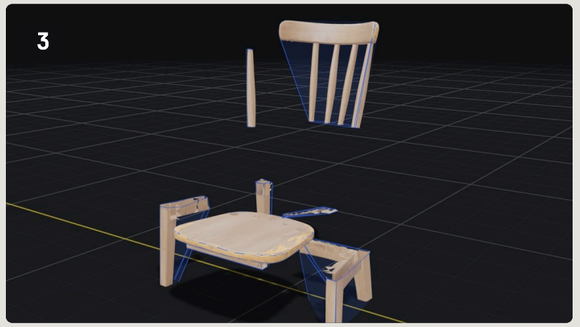
HOW EVERY OBJECT ARRIVES IN THE ARENA



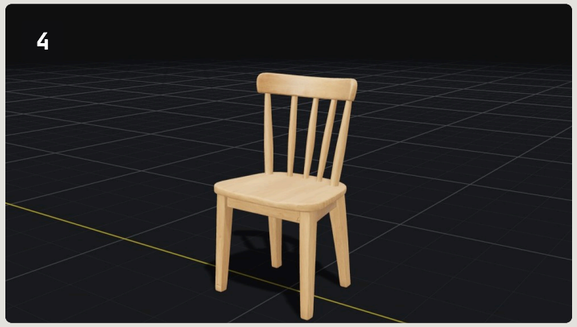
Points condense · Lux3D mesh vertices



Texture sweeps in · real size, in metres



Part-split pulls it apart · hull around each part



Parts close up · ready to drop in

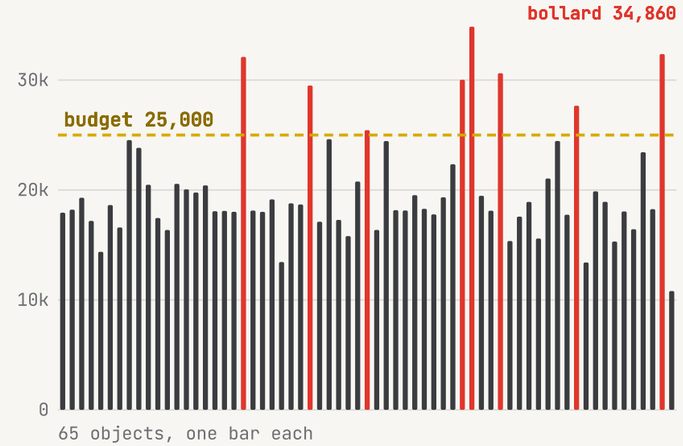
Iteration 3: 305,236 triangles down to 24,250





	First try (photo sneaker)	Now
Request to Lux3D	default (no face limit)	faceCount 20,000
Triangles from Lux3D	305,236	10,804 – 34,860 (median 18,628)
GLB from Lux3D	18.9 MB, two 2048 px PNG textures	7 – 12 MB
After our processing	24,250 triangles, 769 KB	0.3 – 0.8 MB, same triangles
Textures	2 × 2048 px PNG	1 × 1024 px WebP, meshopt-compressed
Over the 25,000 budget	simplified from 305,236	8 objects, simplified to just under 25,000

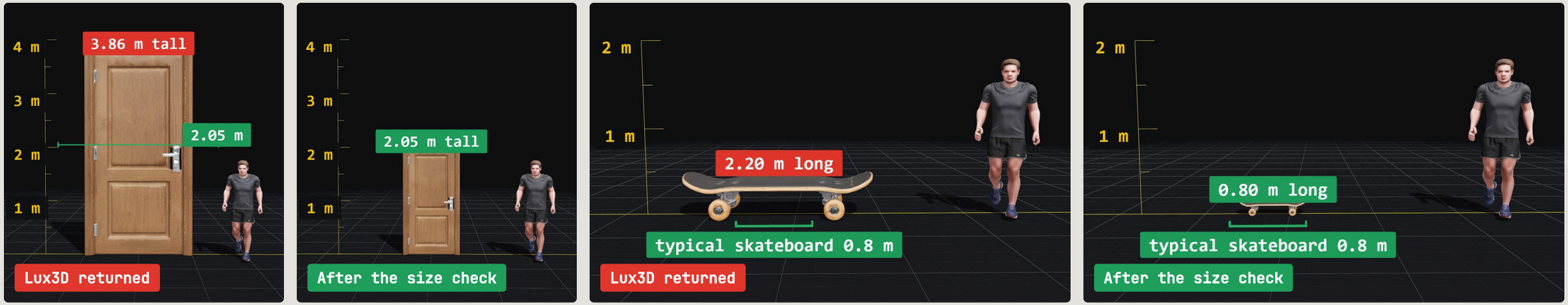
Triangles Lux3D returned, against the budget the browser gets



We ask Lux3D for 20,000 faces up front, so almost everything arrives under the 25,000 budget. The red bars did not and were simplified to just under it.

All 65 objects (people not counted): 614 MB of GLB from Lux3D, 29.3 MB loaded by the browser.

Iteration 4: a 3.86 m door, and the size check



Same meshes, same camera. The person beside each object is the Lux3D-made jogger, 1.8 m; yellow ruler in metres; green marks the typical size.

How it was handled

1. **Compare:** every library object has a typical largest dimension (common retail sizes, in sizes.ts); the measured size of the Lux3D mesh is checked against it.
2. **Rescale:** if the two differ by more than 35%, the mesh is scaled uniformly to the typical size before the hulls and volume are built. Joint pivots of the door scale with it.
3. **Leave alone:** everything else (the other 49 of 60 generations checked) keeps Lux3D's own size. Objects people make themselves have no reference size and keep it too.

Library objects outside the 35% band (largest dimension; 5 of 11 shown)

Object	Lux3D returned	Typical	Scale
Skateboard	2.20 m	0.80 m	× 0.36
Agility ladder	0.60 m	1.50 m	× 2.50
Trash can	0.38 m	0.90 m	× 2.37
Block pile	0.21 m	0.40 m	× 1.90
Door	3.86 m	2.05 m	× 0.53

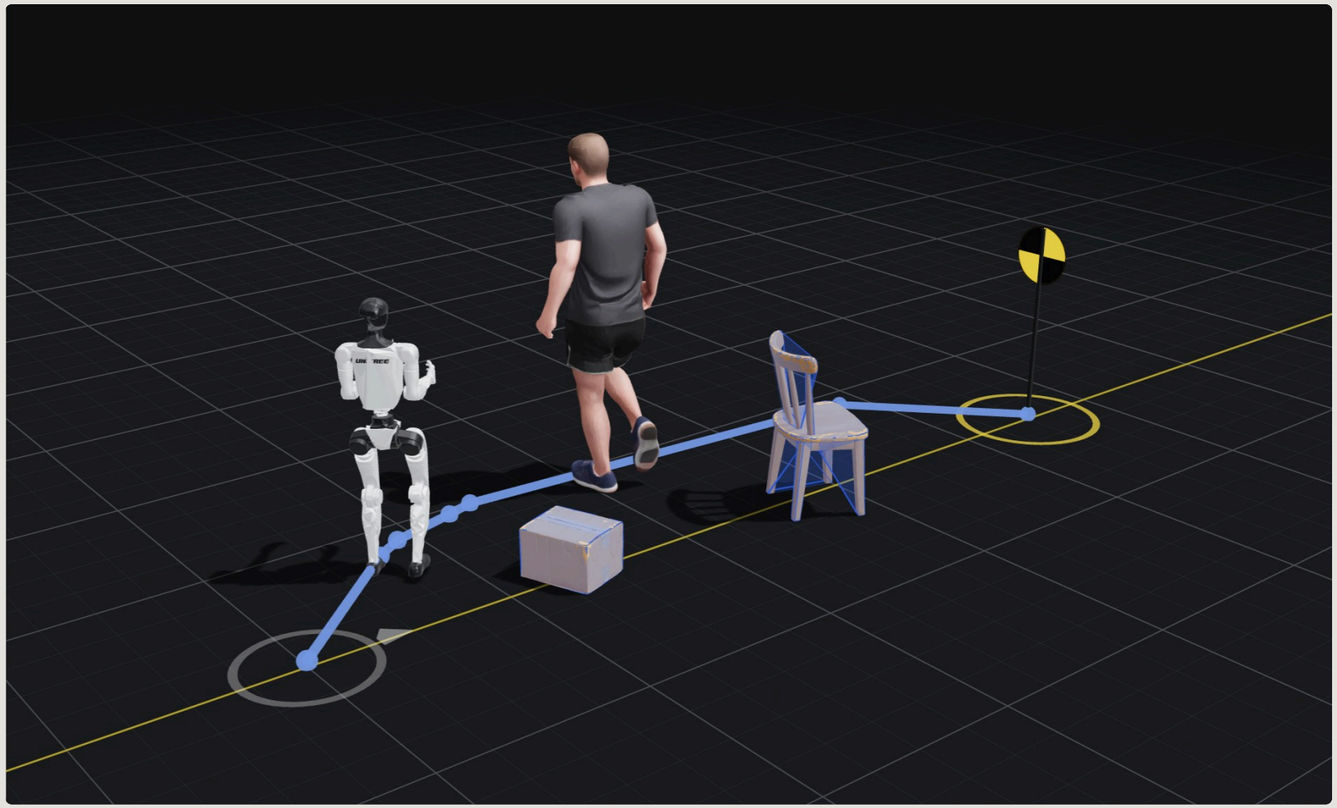
At 400 kg/m³ the unchecked 2.20 m skateboard weighed 58 kg. Volume, and so mass, follows the cube of the scale factor ($0.36^3 = 0.05$).

The robot is real software

The walking policy is the G1 policy Unitree published in `unitree_rl_gym` (BSD-3), the one their sim-to-real example deploys on the physical robot. We did not retrain or change it.

It is blind: it only feels its joints. Planner mode adds a map of object footprints and an A* path, and waits when a walking person is about to cross. The difference between the two pass rates is the value of perception.

Gap to Unitree's Python run after 2 s	**2.5 µm**
Walk 6 s at 0.5 m/s	**2.6983 m (same as Python)**
Speed in the browser	**12× real time (one chair), 3–4× (20 objects)**
10 trials of a chair course	**3.7 s wall time**
Compile a 20-object scene	**163 ms**
One 250 N shove at a random moment and angle	**falls 5 of 10 walks**



Planner mode with collision hulls on. The route (blue) goes around the box and the chair, and the robot has stopped at the start because the jogger is about to cross; the log says “Waiting for someone to cross the path”.

Sense
joint angles and speeds, gravity direction, walking command

›

Unitree policy · 50 Hz
LSTM read from the TorchScript file, plain JavaScript

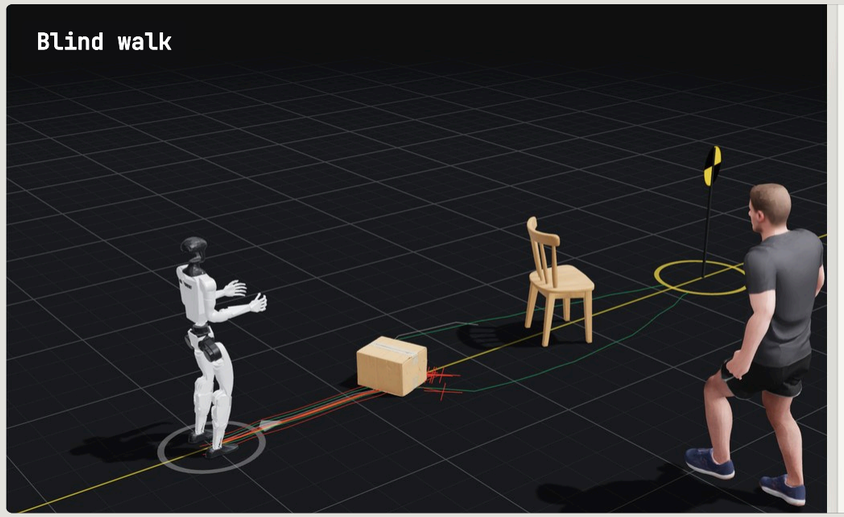
›

PD control · 500 Hz
joint targets into torques

›

MuJoCo · WebAssembly
the robot meets Lux3D part hulls

Ten runs, not one



Test report · blind walk

20% reached the goal

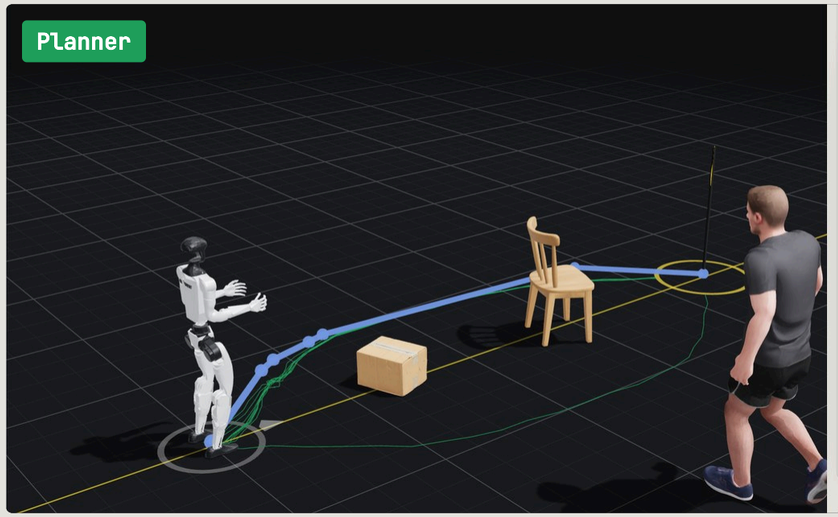
01	02	03	04	05
PASS	FELL	FELL	FELL	PASS
06	07	08	09	10
FELL	FELL	FELL	FELL	TIME

Average time to goal **12.9 s**

Tripped on Cardboard box · 7 times

Click a run to replay it exactly, shove and all.

ROBOT DEFEATED



Test report · planner

100% reached the goal

01	02	03	04	05
PASS	PASS	PASS	PASS	PASS
06	07	08	09	10
PASS	PASS	PASS	PASS	PASS

Average time to goal **15.1 s**

Click a run to replay it exactly, shove and all.

COURSE CLEARED

Pass rate by course

ten seeded runs each; – = not run

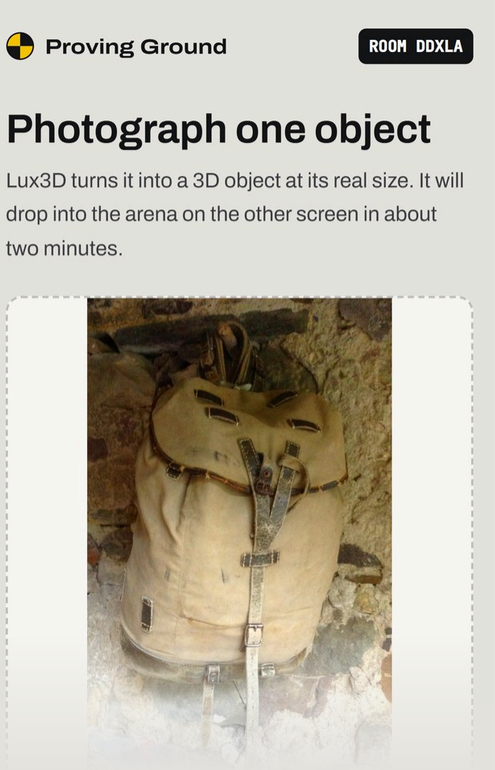
Course	Blind walk	Planner
Starter course, jogger crossing	**2 of 10**	**10 of 10**
Park bench across the path	**0 of 10**	**10 of 10**
Kitchen floor: chair, sneaker, cone	**8 of 10**	–
Same, one 170 N shove per run	**7 of 10**	–
8-part Lux3D chair, trip test (1.4 m, chair turned and shifted each run)	**5 of 10**	–
0.55 kg sneaker; 2.0 kg cone	**10 of 10 each**	–
Fixed 6 cm or 12 cm step	**0 of 10**	–

With the jogger crossing, a blind walk mostly trips on the cardboard box and reaches the flag 2 times in 10. The planner routes around the objects and waits for the jogger to pass: 10 of 10 in the arena, 10 of 10 in the check script (scripts/jogger-check.mjs). That gap is what perception is worth, measured.

A Lux3D park bench across the path is the clearest case: the blind walk pushes a 56 kg bench and stalls; the planner routes around it. In the 8-part chair trip test, where the chair is turned and shifted at random each run (harder than the square-on comparison on the hull slide), the blind walk gets 1 fall and 4 stuck pushing it.

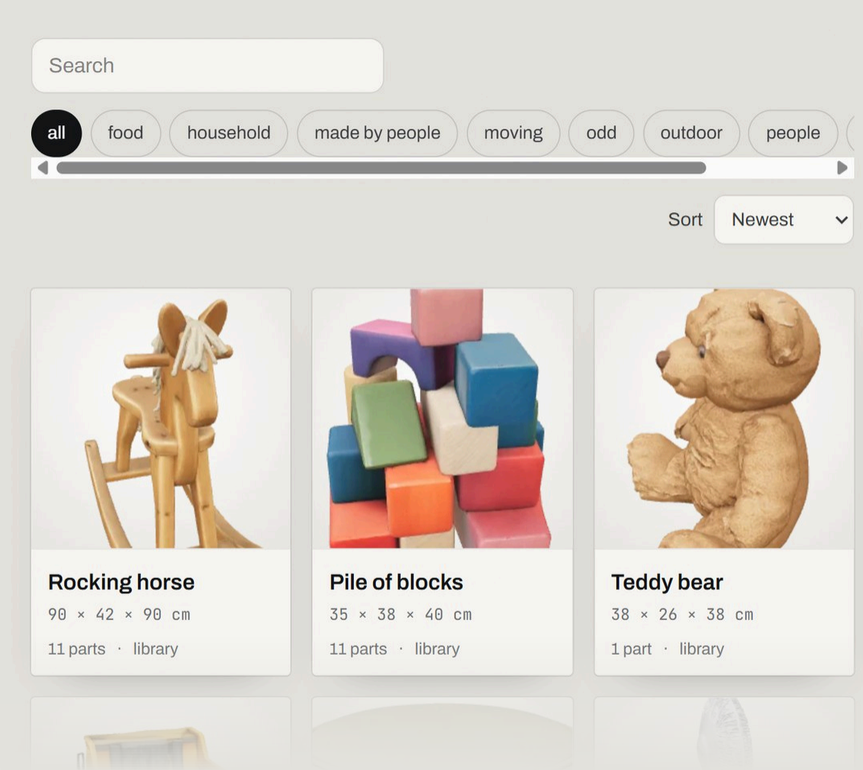
Fixed steps trip the robot every time, and every trial records what it tripped on.

Built for people to play with



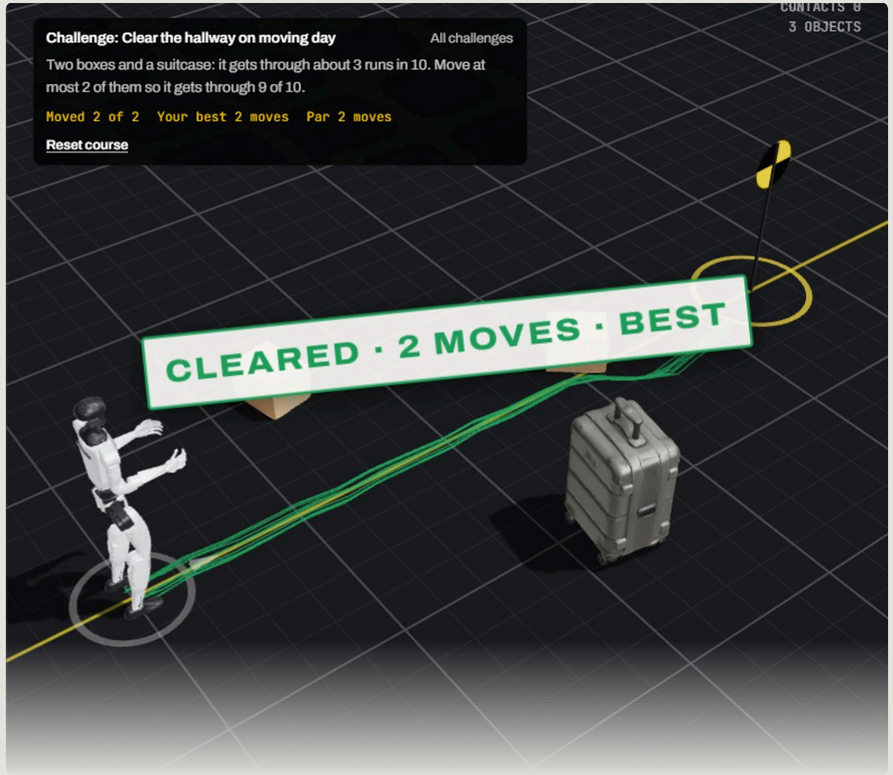
Snap from your phone

Scan the QR code, photograph something real, and it lands in the desktop arena. The phone page sends nothing until you press the button.



A shared library

Every object anyone makes, with parts and hulls, downloadable as a robot-ready pack: textured GLB, one STL hull per part, MJCF body with masses.



Challenges and the gauntlet

A goal, a limit and a score: fix a course in as few moves as possible, trip the robot with the least mass, or steer it home against the clock. Plus 23 courses ranked by how often the robot fails them.

What it cannot do yet, and what is next

LIMITS

- The controller is blind and was trained on flat ground; it never sees the objects. Planner mode adds a map of footprints, not vision.
- Each part collides as its convex hull; hollow parts (a bowl) are filled.
- Mass is an estimate from volume and a density you can change.
- Walking people are kinematic: they push the robot but cannot be pushed.
- One Lux3D generation at a time per account, so new objects can queue.
- Lux3D sometimes misjudges size. The size check covers library objects only; things people make keep Lux3D's own size.
- Trial results are run and reported by each visitor's browser; anyone can rerun a course to check it.

NEXT

- Grasping: the same objects for a manipulation policy (Lux3D part-split gives handles and lids for free).
- Lux3D world generation as the floor plan: test inside a generated room, not on an empty floor.
- Export courses as MuJoCo and Isaac scenes for CI on real robot code.

Scan to open the arena

Try it: **proving-ground-snowy.vercel.app**

Source: github.com/RohanGlitched/proving-ground

Video: https://youtu.be/6pRu4K838hk

BUILT ON

Aholo Lux3D
Manycore Tech: 3D generation, part-split, rig, walk cycle, pricing

Unitree G1
model and the unitree_rl_gym walking policy, BSD-3-Clause

MuJoCo
Google DeepMind, Apache-2.0, compiled to WebAssembly

three.js
rendering, MIT